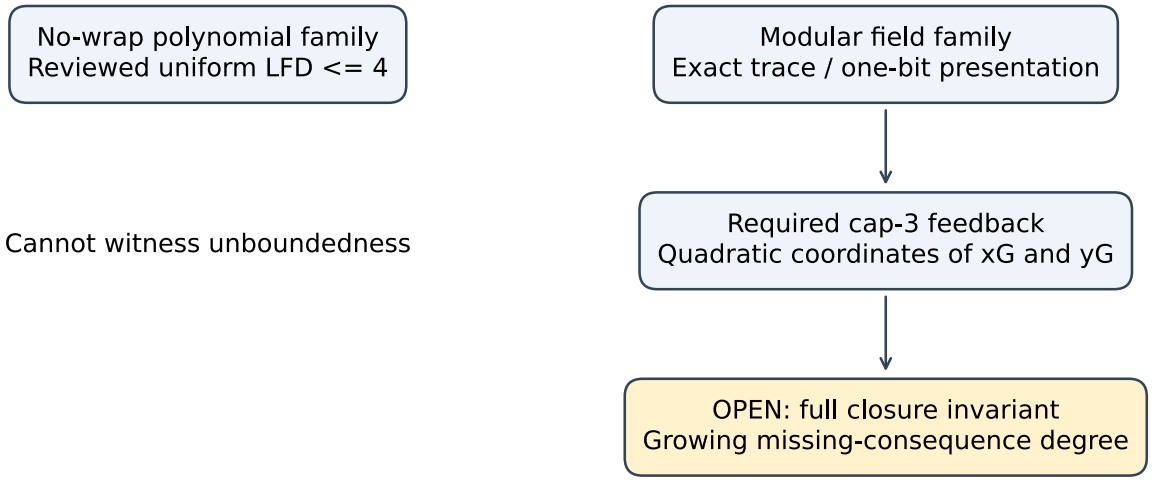
Fixed-m=2 modular proof frontier

2026-11-09. Symbolic derivations and independent written reviews.



The no-wrap polynomial family has a reviewed uniform last-fall bound of four. It cannot supply an unboundedness example. For the modular half-diagonal family, the exact one-bit presentation G has quadratic consequences xG and yG derivable at cap three, because field squaring is Boolean-linear.

Trace-output cancellation conditions depend on the actual multiplication tensor. Trace elimination may add syzygies; all subsequent feedback must be included. No stable full-closure obstruction or growing lower bound has been proved.

A separate selector lemma gives LFD at most N+1 for an exactly reconstructed normalized N-variable Boolean system, preserving nonlinear maps and code multiplicity. It is not a growing lower bound or dataset admission.

Evidence: modular-degree-three-field-falls.md and review; finite-selector-bounds-and-transfer.md and review; prior no-wrap-uniform-cap-four.md and review.

No new trials, solver calls, native builds or cloud jobs. Existing trial total:13. These are written proofs, not formal-kernel certification. Next: full trace-slice closure classification and a missing-consequence construction at growing degree.

Frobenius follow-up: Simultaneous twists stay in the existing span; nonzero relative twists do not give automatic field-valued quadratic falls. For coprime shifts, scalar trace cancellation forces inverse-power invariance of the nonzero subspace elements. This necessary condition precedes trace elimination; full closure remains open. Evidence: modular-frobenius-multiplier-obstruction.md and independent review.